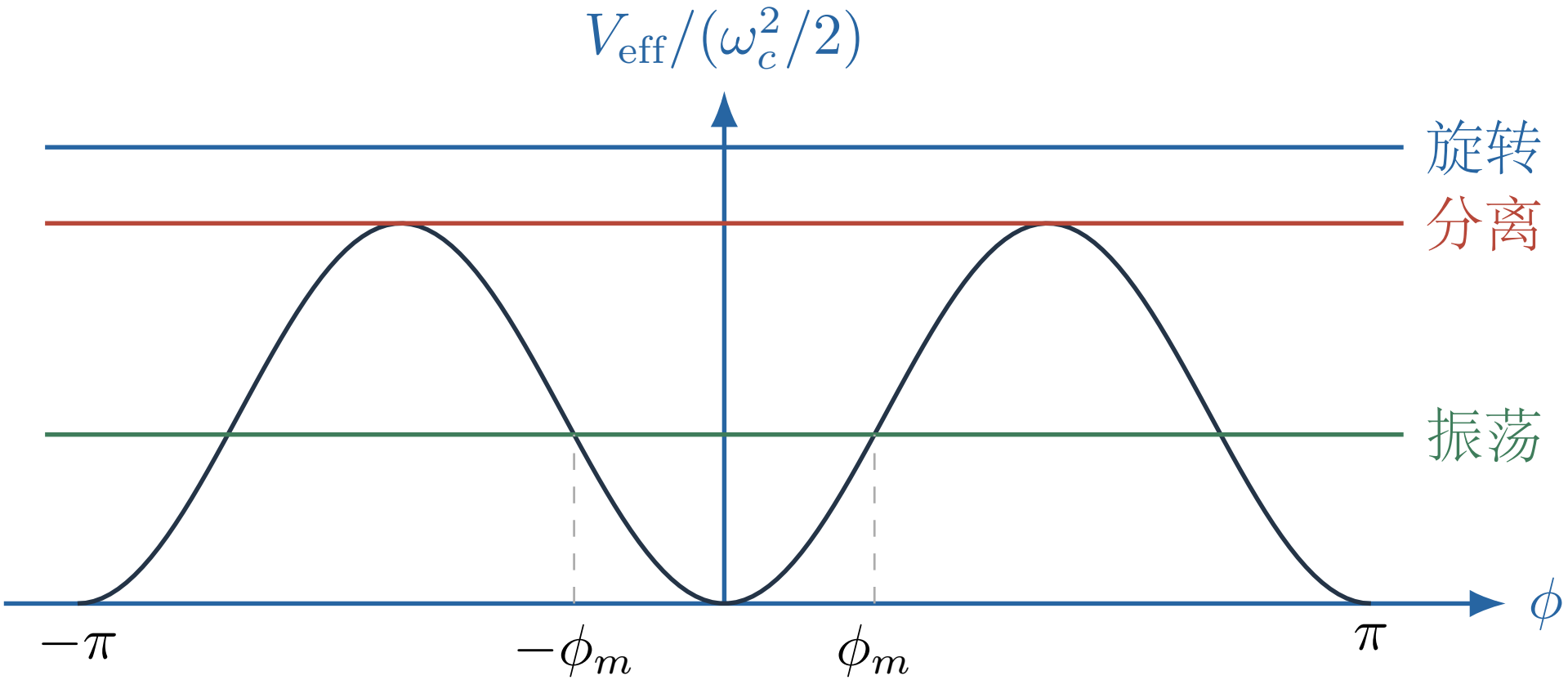

$$\omega_z = 0\colon\ V_{\text{eff}} = \tfrac{1}{2}\omega_c^2 \sin^2\phi$$